



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 • STANDARD 6.GR.1

# Area of Triangles

Lesson 5-3 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can find the area of a triangle using the formula  $A = \frac{1}{2} \times \text{base} \times \text{height}$ .

**Language Objective:** I can explain my steps using the words base, height, area, and perpendicular.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Base

Height

Area

Perpendicular

Composite figure

Formula



Tap any word to see what it means and a picture.

1

The side of a triangle used as the bottom for measuring area is the

2

The perpendicular distance from the base to the opposite vertex is the

3

The amount of flat space inside a triangle is its .

4

Two lines that meet at a  $90^\circ$  angle are .

5

A shape made of two or more basic shapes combined is a

- 6 A math rule written with symbols, like  $A = \frac{1}{2} \times b \times h$ , is a

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**On the grid your triangle (base 12, height 8) fits inside a 12 by 8 rectangle. Why do we divide by 2 to find the triangle's area?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Base** — A side of the triangle you use to find the area.

*Base — Un lado del triángulo que usas para hallar el área.*

- ☐ **Height** — The straight-up distance from the base to the top corner.

*Altura — La distancia recta desde la base hasta la esquina de arriba.*

- ☐ **Area** — How much space is inside a flat shape.

*Área — Cuánto espacio hay dentro de una figura plana.*

- ☐ **Perpendicular** — Two lines that meet to make a square corner (90 degrees).

*Perpendicular — Dos líneas que se unen formando una esquina recta (90 grados).*

- ☐ **Formula** — A math rule written with symbols.

*Fórmula — Una regla matemática escrita con símbolos.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**A triangle is half of a \_\_\_\_ with the same base and height.**

*Un triángulo es la mitad de un \_\_\_\_ con la misma base y altura.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The height is the perpendicular distance from the base to the top corner.

**Evidence:** Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

**Reasoning:** This evidence connects the definition of area to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **A triangle is half of a \_\_\_\_ with the same base and height.**

*Un triángulo es la mitad de un \_\_\_\_ con la misma base y altura.*

- **So the area is  $1/2$  times 12 times 8, which equals \_\_\_\_.**

*Entonces el área es  $1/2$  por 12 por 8, que es \_\_\_\_.*

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Base
2. **Notes 2:** Height
3. **Notes 3:** Area
4. **Notes 4:** Perpendicular
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **Watch for this mistake:** *A common mistake in Area of Triangles is forgetting the  $\frac{1}{2}$  — students multiply base  $\times$  height like a rectangle and stop there, or they use the triangle's slanted side instead of the perpendicular height that's already given. Always check: did I take half, and did I use the straight-up height, not the slanted side?*
8. **Try It 1:** A. 14 sq ft —  $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \times 4 = \frac{1}{2} \times 28 = 14 \text{ sq ft}$ . Multiply base  $\times$  height first (28), then take half (14).
9. **Try It 2:** A. 33 sq ft —  $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 11 \times 6 = \frac{1}{2} \times 66 = 33 \text{ sq ft}$ .